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c553857 · 1 minute ago

 LICENSE	Initial commit	5 years ago
 LU.py	final	3 years ago
 README.md	succes	1 minute ago
 Screenshot 2026-05-24 142...	succes	1 minute ago
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AIM:

Equipments Required:

- ## Algorithm

Step 2: Prepare the lists from each linear equations and assign in np.array()

Step 3: Using scipy.linalg import lu ,we can find L and U matix.

Step 4: By using lu_factor ,lu_solve .we can find lu decomposition matrix.

Step 5: End the program

Program:

(i) To find the L and U matrix

```
'''Program to find L and U matrix using LU decomposition.
Developed by: Balaji B
RegisterNumber: 212225040040'''
import os
os.environ['OPENBLAS_NUM_THREADS']="1"
import numpy as np
from scipy.linalg import lu
a=np.array(eval(input()))
piv,Lmatrix,Umatrix=lu(a)
print(Lmatrix)
print(Umatrix)
```



(ii) To find the LU Decomposition of a matrix

```
'''Program to solve a matrix using LU decomposition.
Developed by: Balaji B
RegisterNumber: 212225040040
'''
import os
os.environ['OPENBLAS_NUM_THREADS']="1"
import numpy as np
from scipy.linalg import lu_factor,lu_solve
a=np.array(eval(input()))
b=np.array(eval(input()))
piv,lu=lu_factor(a)
result=lu_solve((piv,lu),b)
print(result)
```



Output:

Use LU Decomposition to find L and U matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[1, 0, 0,] [1, 1, 0,] [0.6666667 0.8333333 1,]] [[3, 2, 7,] [0, 2, -6,] [0, 0, 1.3333333]]

Answer: (penalty regime: 0 %)

Reset answer

```
1 '''Program to find L and U matrix using LU decomposition.
2 Developed by: Balaji B
3 RegisterNumber: 212225040040'''
4 import os
5 os.environ['OPENBLAS_NUM_THREADS']='1'
6 import numpy as np
7 from scipy.linalg import lu
8 a=np.array(eval(input()))
9 piv,latrix,u=lu(a)
10 print(latrix)
11 print(uatrix)
```

Input	Expected	Got	
✓ [[3, 2, 7], [2, 3, 1], [3, 4, 1]]	[[1, 0, 0,] [1, 1, 0,] [0.6666667 0.8333333 1,]] [[3, 2, 7,] [0, 2, -6,] [0, 0, 1.3333333]]	[[1, 0, 0,] [1, 1, 0,] [0.6666667 0.8333333 1,]] [[3, 2, 7,] [0, 2, -6,] [0, 0, 1.3333333]]	✓
✓ [[3, 1, 8], [4, 5, 7], [8, 9, 1]]	[[1, 0, 0,] [0.625 1, 0,] [0.5 -0.10810811 1,]] [[0, 0, 1,] [0, -4.625 7.375] [0, 0, 7.2972973]]	[[1, 0, 0,] [0.625 1, 0,] [0.5 -0.10810811 1,]] [[0, 0, 1,] [0, -4.625 7.375] [0, 0, 7.2972973]]	✓

Passed all tests! ✓

Use LU Decomposition to solve a matrix.

For example:

Input	Result
[[3, 2, 7], [2, 3, 1], [3, 4, 1]] [4, 5, 7]	[0.875 1.125 -0.125]

Answer: (penalty regime: 0 %)

Reset answer

```
1 '''Program to solve a matrix using LU decomposition.
2 Developed by: Balaji B
3 RegisterNumber: 212225040040
4 ...
5 import os
6 os.environ['OPENBLAS_NUM_THREADS']='1'
7 import numpy as np
8 from scipy.linalg import lu_factor,lu_solve
9 a=np.array(eval(input()))
10 b=np.array(eval(input()))
11 l,u,p=lu_factor(a)
12 x=lu_solve(l,u,b,p)
```

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Languages

- Python 100.0%

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Based on your tech stack



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[Configure](#)

Python Package using Anaconda

Create and test a Python package on multiple Python versions using Anaconda for package management.

[Configure](#)

SLSA Generic generator

Generate SLSA3 provenance for your existing release workflows

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